



Enterprise Asset Lifecycle (EAMS) v15

Asset Lifecycle Module

End User Documentation

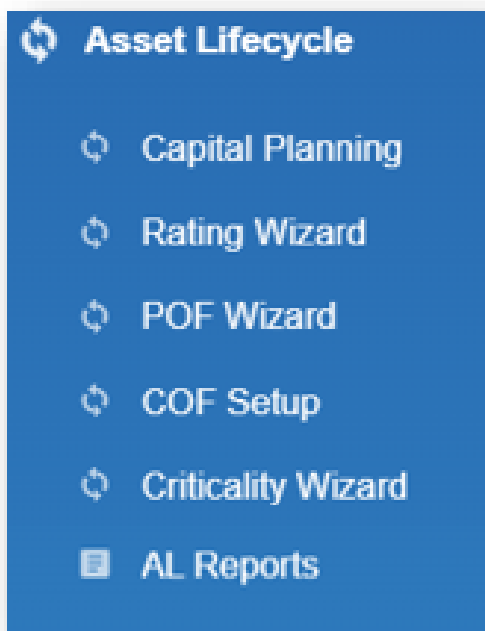
Update 9/25/2024

Asset Lifecycle Module

The Asset Lifecycle Module is a paid/licensed module composed of a variety of tools to assist organizations with the decision-making process. By combining asset data and work order data, these tools can be leveraged to help planners determine asset lifecycles, replacement schedules, budgeting and overall asset performance.

Asset Lifecycle Tools

With the potential of many assets, it may be prohibitive to perform individual assessments and replacement schedules on every asset within an organization(s). As part of the Asset Lifecycle module, MaintStar provides tools to assist with the ratings, assessments, replacement schedules and budget development. These tools include:



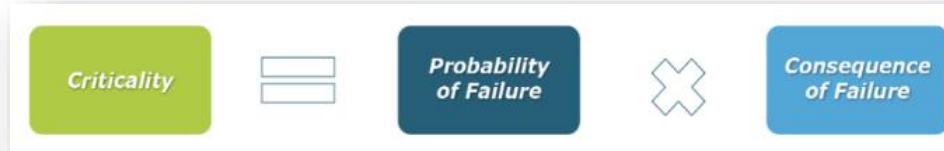
- **Rating Wizard:** The Rating Wizard is a tool that automates the ratings of assets based on life, maintenance and use data within MaintStar. Ratings can be scale(s) of an organization choosing. For instance, there are Pavement Condition Indexes that have rating scores from 0-100; a simple 1 to 5 scale, an A-F scale or a combination of all. Using factors such as age, cost to maintain and meters (for assets with meters), automated ratings can be applied to assets. This is the best way to apply real world logic and information to rate assets according to thresholds set by an organization.

- **COF Setup:** The Consequence of Failure (COF) Setup tool provides an easy way to apply or update COF values to Asset Types. COF values help to identify the value of assets to organizations based on a 1-5 scale with a 1

depicting a low impact if an asset fails and a 5 depicting a high or significant impact.

- **POF Wizard:** The Probability of Failure (POF) Wizard provides an easy way to apply or update POF values to assets. POF values help to identify the likelihood of an asset's failure. Using factors such as age, cost to maintain and meters (for assets with meters), automated POF scores can be applied to assets. POF ratings are based on a 1-5 scale with a 1 depicting a low likelihood of an asset's failure and a 5 depicting a high likelihood of failure.

- **Criticality Wizard:** The Criticality Wizard is the result of the COF and POF ratings. Calculating Criticality is a multiplication of POF x COF which is a standard calculation for determining an organization's critical assets.



- **Capital Planning:** The Capital Planning tool is a separate license from Asset Lifecycle and provides the ability to 'model' replacement dates and costs base one, some or all the factors above, such as age, meter, cost, ratings, usage and assessments. Going beyond a fixed or static replacement date, this is particularly helpful for budgeting for current and/or future year replacements based on the factors above.
- **Asset Lifecycle (AL) Reports:** Standard reports for all of the above modules. As with most modules in MaintStar Reports and Queries can also be used to meet an organization's unique needs.

Factors Affecting Asset Life

Whether an asset is a hydrant, pipe, vehicle, generator or building, it has a theoretical end of life date. This date can be shortened or extended by a variety of factors to include frequency of use, quality of maintenance, criticality of the asset and overall asset condition. The number of assets that an organization is responsible for maintaining may preclude rating or evaluating every single asset. Common factors that MaintStar uses to evaluate an asset include:

Asset Factor	Description	Where to Set
EOL Age: expected life versus current age	This calculates the age of the asset using: Install Date – Current Date = Current Age An additional calculation determines years remaining: EOL Age – Current Age = Life Remaining	End of Life (EOL) Age Set at Asset Type Level
EOL Maintenance Cost: a percentage of allowable maintenance cost versus replacement cost	End of Life (EOL) Cost is set at Asset Type Level as a percentage. For instance, setting a value of 50% would be factored into the tools listed above.	The EOL Cost threshold is set to individual Asset Type(s).

	EOL Cost – Total Corrective Work Order Cost = WO Cost %	Work Order Cost is automatically calculated by MaintStar through Work Order cost tracking.
EOL Meters (mileage or hours): expected meter life versus actual reading	EOL Meters is set at the Asset Type level and helps to report whether an asset equipped with a meter is within or outside desired usage thresholds by calculating: EOL Meter – Current Meter = Meter Life %* <i>*Meter types are set at the asset type level and MaintStar supports multiple meters, but only the default meter is considered for this calculation</i>	End of Life (EOL) Meter Set at Asset Type Level
Ratings: an 'excellent to bad' scale that allows for simple asset ratings	Ratings can be customized by any organization and are typically from "excellent to bad" in a numbered or lettered format, ie "1-5" or "A-F". The scales are called "Condition Indexes". These are applied to individual Assets by an individual.* <i>*The Condition Index table stores the values for the ratings.</i>	Condition Indexes must exist in Setup Tables and then must be applied to the Asset Type. Once setup, Ratings can be individually recorded through an asset's "Assessment" tab. Alternatively, the Rating Wizard can be used to automate these ratings.
Assessments: a formal process to rate assets with replacement cost and/or life remaining	Using Condition Indexes and a formal assessment process, the Asset Assessment tool can be used provided updated replacement and/or cost information to assets. These are applied to individual Assets by an individual.	Once setup, Assessments can be individually recorded through an asset's "Assessment" tab.

TO ENSURE USABLE OUTPUTS, ORGANIZATIONS MUST CREATE AND MAINTAIN THEIR DATA IN ACCORDANCE WITH THIS DOCUMENT'S RECOMMENDATION

Configuration and Uses of End of Life (EOL) Settings

To ensure accurate use of the tools above, there are settings that must be applied to Assets and Asset Types. Below are those critical settings:

Using End of Life (EOL Age) Settings

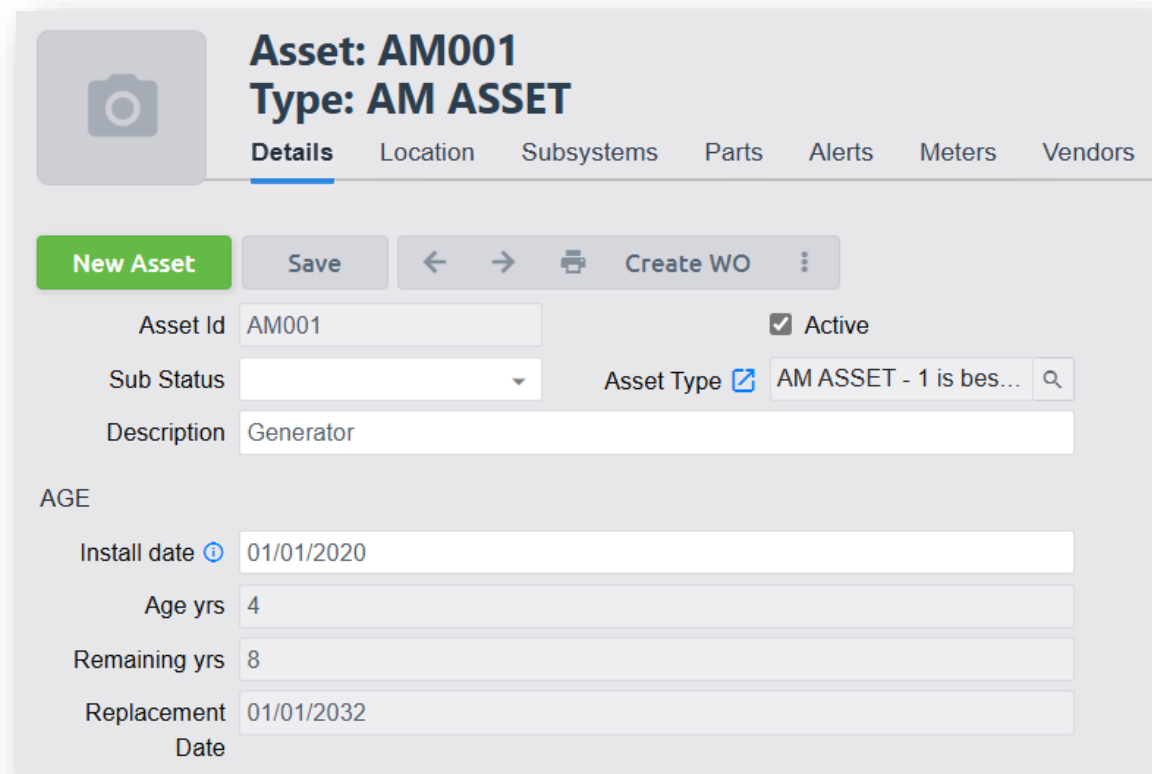
EOL Age: As a reminder, this is set at an Asset Type by Asset Type basis. Therefore, "Heavy Duty" and "Light Duty" trucks can be given different life expectancies.

1. Navigate to Assets > Asset Type and locate the desired Asset Type. There is a field labeled “EOL Age”. Enter the desired life expectancy in years, click Save.

A screenshot of a web form showing a field labeled "EOL Age Years" with an information icon (i) and a text input field containing the number "12".

Install Date: This field is on the Asset itself, and this date typically refers to the date the asset has been purchased and/or placed into service.

1. Navigate to Assets and locate the desired Asset. At an individual asset basis, the field labeled “Install Date” (field name = install_date) must be populated. Following is an example of the calculation at the Asset level that shows the asset with a 12 year life expectancy and the number of remaining years based on the Asset Type setting:

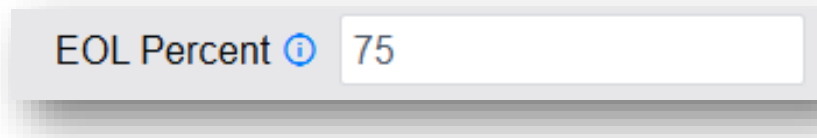
A screenshot of the "Asset: AM001" details page in the MAINTSTAR system. The page shows the asset's details, including its ID (AM001), status (Active), and description (Generator). It also displays the "AGE" section with fields for "Install date" (01/01/2020), "Age yrs" (4), "Remaining yrs" (8), and "Replacement Date" (01/01/2032). The page includes navigation tabs for Details, Location, Subsystems, Parts, Alerts, Meters, and Vendors, and action buttons like "New Asset", "Save", "Create WO", and "Create WO".

*The field may or may not be present on every asset screen customization. If it is not present, customize the screen and add the “install_date” field.

Using End of Life Cost (EOL Percent)

EOL Percent: As a reminder, this is set at an Asset Type by Asset Type basis. Therefore, differing cost values can be placed to differing asset types. The purpose of this field is to provide a Work Order Cost threshold for all corrective (not planned) work completed against an asset.

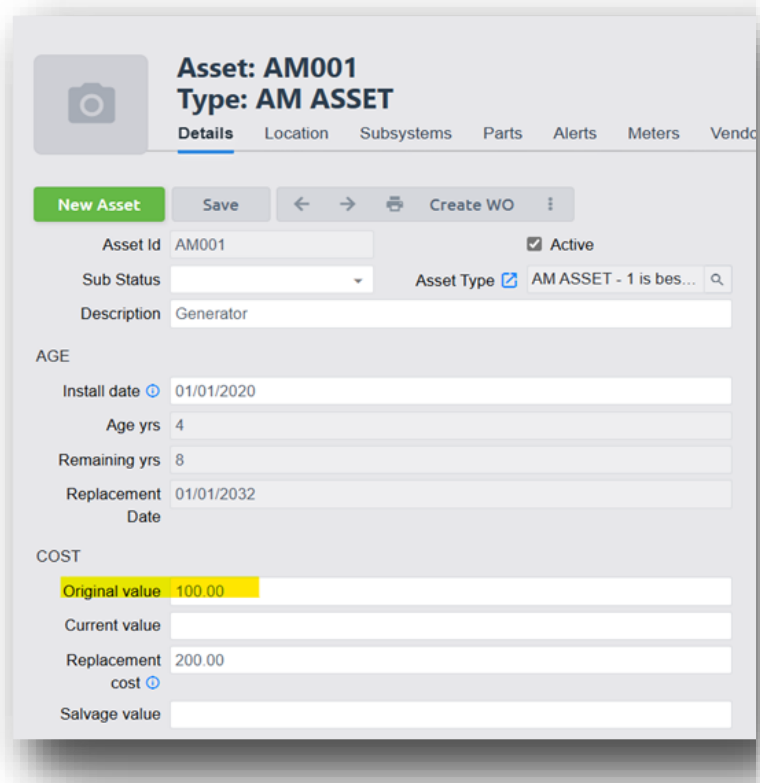
1. Navigate to Assets > Asset Type and locate the desire Asset Type.
2. To set (or confirm) the Asset Types WO Cost, locate the field labeled “EOL Percent”. Set the desired value and click Save.*



Original and Replacement Cost: for the tools above to work correctly, original and replacement costs should be set. If this data does not exist the tools will use other factors as best it can to assess ratings.

1. At an individual asset basis, the field labeled “Original Value” (field name = original_value) must be populated with original cost data.*
2. At an individual asset basis, the field labeled “Replacement Value” (field name = replacement_cost) should be populated with replacement cost data, even if it is an estimate.*

*The field may or may not be present on every asset screen customization. If it is not present, customize the screen and add the “original_value” field.

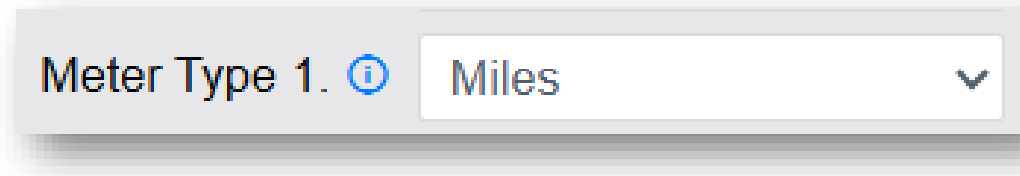


3. The example above has a EOL Cost of 75% and an Original Value of \$100. If Work Order Cost exceeds \$75 then the asset is to be considered more expensive to repair then replace.

Using End of Meter (EOL Meter)

EOL Meter (if meters are used): As a reminder, this is set at an Asset Type by Asset Type basis. Therefore, “Heavy Duty” and “Light Duty” trucks can be given different meter values such as 200,000 miles for heavy duty and 150,000 miles for light trucks.

1. Navigate to Assets > Asset Type and locate the desire Asset Type.
2. To set (or confirm) the Asset Types default meter, locate the field labeled “Meter 1 Type”. Set the meter and click Save.



*If there are NO meters set in the meter type, they must be set up by an administrator under Setup > Tables > Meter Types.

*If the assets existed before the meter type was set, then it may be necessary to click the “Update Asset Meter” link which retroactively places the meter (with 0 mileage) on all assets within the type.

Updating Meters

Meter data can be obtained through a variety of means, all of which are extensively covered in documentation. In summary, meters can be updated via Work Orders, manually on Assets, and/or through a fuel system integration.

Configuration and Uses of Condition Indexes and Ratings

To use asset ratings, condition indexes must be configured (by an administrator or with the help of MaintStar). The ratings are a way to document an assets condition by assigning a value. Traditionally different types of assets can be rated with different scales. Some may rate on a 5-point scale, 1 to 5 with 5 being best. Some use a 10-point scale where 1 is the best and 10 is the worst condition.

Condition Index: the name of the scale, ie “1 to 5” or “PCI” for road pavement ratings. MaintStar supports a multitude of industry standard or custom rating scales. When setting up the index, a best and worst score value must be set. Condition Indexes are configured within Setup > Tables > Condition Index. In the “1 to 10” example below, the best and worst ratings are set.

Tables

Table list:

Condition Index 

Condition Indexes, such as PCI, which are used to measure the condition of

 Add				
☐ Edit	Condition Index	Description	Best Rating	Worst Rating
☐ Edit	1 TO 10	1 is best, 10 is worst.	1.0	10.0

Condition Rating: contains the values that apply to the index. The example below shows the individual rating values assigned to the “1 to 10” index.

Tables

Table list:

Condition Rating 

Possible values for Condition Rating of an Asset, for a given Condition Index

 Add				
☐ Edit	Condition Index	Condition Rating	Description	End of Life
☐ Edit	1 TO 10	1.0	Brand-new	☐
☐ Edit	1 TO 10	2.0	Excellent	☐
☐ Edit	1 TO 10	3.0	Very Good	☐
☐ Edit	1 TO 10	4.0	Good	☐
☐ Edit	1 TO 10	5.0	Fair to Good	☐
☐ Edit	1 TO 10	6.0	Fair	☐
☐ Edit	1 TO 10	7.0	Fair to Poor	☐
☐ Edit	1 TO 10	8.0	Poor	☒
☐ Edit	1 TO 10	9.0	Very Poor	☒
☐ Edit	1 TO 10	10.0	Failed	☒

Setting POF Thresholds

In addition to setting up Condition Ratings, it may be helpful to assign POF thresholds, per the example below. This is used by all of the tools in the Asset Lifecycle module. There are two critical fields, “End of Life and “POF” in this table:

- **End of Life:** placing a checkbox in this field(s) signifies what an asset's end of life is based on it's condition. For example, a rating of "10" represents "Failed" so the organization may consider this End of Life for Asset Lifecycle and Reporting purposes
- **POF:** allows an organization to place POF values to corresponding Condition Rating. For instance, if a perfect rating is "1" then assign a POF of "1", and a rating of "10" indicates a POF of 5, imminent failure, and so on.

Tables

Table list:

Condition Rating 

Possible values for Condition Rating of an Asset, for a given Condition Index.

 Add

	1 to 10				
☐ Edit	Condition Index	Condition Rating	Description	End of Life	POF
☐ Edit	1 TO 10	1.0	Brand-new	☐	1
☐ Edit	1 TO 10	2.0	Excellent	☐	2
☐ Edit	1 TO 10	3.0	Very Good	☐	2
☐ Edit	1 TO 10	4.0	Good	☐	3
☐ Edit	1 TO 10	5.0	Fair to Good	☐	3
☐ Edit	1 TO 10	6.0	Fair	☐	3
☐ Edit	1 TO 10	7.0	Fair to Poor	☐	4
☐ Edit	1 TO 10	8.0	Poor	☐	4
☐ Edit	1 TO 10	9.0	Very Poor	☒	5
☐ Edit	1 TO 10	10.0	Failed	☒	5

Assigning a Condition Index to an Asset Type: In order to use the condition ratings on assets, the indexes MUST be applied to the corresponding Asset Type. MaintStar supports multiple indexes on Asset Types, but only one can be a default (important to know for reports). Navigate to Assets > Asset Type > locate the appropriate Asset Type and click on "Condition Indexes". From here the appropriate index and default value can be added or updated.

 Asset Type:: AM ASSET						
Details	Attachments	Asset Subtype	Subsystem	Condition Indexes	Assessments	LTD Meter Update
 Add						
 Edit	Asset Type	Condition Index	Default	Description	Best Rating	Worst Rating
 Edit	AM ASSET	1 TO 5	☒	1 is best, 5 is worst	1.0	5.0

Using Asset Assessments

Asset Assessments are a tool MaintStar provides to record rating and replacement information to individual assets. This is often done as part of a planned project to evaluate assets. Assessments can then be recorded in MaintStar to show an assets history of ratings. To use assessments, corresponding indexes and ratings must be applied (see above).

Assessments are sometimes completed by contracted professional service providers to view and analyze the conditions of assets and provide reports of findings. It is critical that assessors use the same identification of assets as they are in exist in MaintStar. This is important as that information will need to be entered into MaintStar to become part of each assets record.

Assessments can be found in Assets > Assets > Assessment Tab. Following is an example:

Asset: AM001

Type: AM ASSET

Details

Location

Subsystems

Parts

Alerts

Meters

Vendors

New Asset

Save

←

→

Create WO

⋮

Asset Id

AM001

☒ Active

Sub Status

Asset Type

☒ AM ASSET - 1 is bes...

Description

Generator

AGE

Install date

01/01/2020

Age yrs

4

Remaining yrs

8

Replacement Date

01/01/2032

COST

Original value

100.00

Current value

Replacement cost

200.00

Salvage value

RATINGS

☐ Rating type

Rating 5.00

▼

Asset: AM001

Type: AM ASSET

Details

Location

Subsystems

Parts

Alerts

Meters

Vendors

Warranties

History

PM

Assessments

»

Assessments

Add

☐ Edit	Date	Assessed By	Condition Index	Condition Rating	Condition Descr	Replacement Cost	Additional Cost	Years Re
☐ Edit	01/01/2020	admin	1 TO 5	1.0	Excellent			
☐ Edit	07/22/2024	admin	1 TO 5	5.0	Failed			

Install Date: 01/01/2020

Mfg Estimated Lifespan: 12

Original Replacement Date: 01/01/2032

Last Assessment: 07/22/2024

Assessed Remaining Life: 12

Assessment Replacement Date: 07/22/2036

The assessment screen has many different fields to record data. The key fields would be:

- Assessment Date
- Replacement cost in today's dollars
- Years of life remaining
- There are many other useful values such as rating, criticality, and others, but the replacement cost and years of life remaining are the minimum values required

When a rating score is recorded on this screen, it will update the asset rating with the current score on the main Asset Page to provide a quick glance of ratings.

Benefits of Using Assessments and Asset Ratings

Over time, capturing ratings and assessments can help an organization track an assets overall condition over time. With this information an assets degradation or improvement can be seen. The example below shows a road with Pavement Condition Index (PCI) ratings over time.



Asset: 101
Type: MAINTAINED ROAD

DetailsTraffic CountsLocationAlertsPMAssessments

Assessments

+

Add

☐	Date	Assessed By	Condition Index	Condition Rating	Condition Descr
☐	01/02/2022	James.Despain	PCI	50.0	Fair
☐	01/02/2021	James.Despain	PCI	60.0	Fair
☐	01/02/2020	James.Despain	PCI	60.0	Fair
☐	01/02/2019	James.Despain	PCI	60.0	Fair
☐	01/02/2018	James.Despain	PCI	60.0	Fair
☐	01/02/2017	James.Despain	PCI	60.0	Fair
☐	01/02/2016	James.Despain	PCI	60.0	Fair
☐	01/02/2015	James.Despain	PCI	60.0	Fair
☐	01/02/2014	James.Despain	PCI	60.0	Fair
☐	01/02/2013	James.Despain	PCI	80.0	Good condition
☐	01/02/2012	James.Despain	PCI	80.0	Good condition
☐	01/02/2011	James.Despain	PCI	50.0	Fair
☐	01/02/2010	James.Despain	PCI	50.0	Fair
☐	01/02/2009	James.Despain	PCI	60.0	Fair
☐	01/02/2008	James.Despain	PCI	70.0	Good

Putting it all Together

The above has described the purpose of the Asset Lifecycle Module, the intention of the various tools, and the configuration required within MaintStar to make it all work. With this understanding, the various tools can now be effectively used.

Using the Rating Wizard

Recognizing the difficulty in manually evaluating every single asset, MaintStar provides a tool named “Asset Rating Wizard” that provides a ‘mass’ rating to assets based on the End of Life (EOL) data, as mentioned at the beginning of this document. This module can be accessed from Asset Lifecycle > Asset Rating Wizard. Given that this tool can perform bulk updates to assets, it is important that only trained and authorized individuals use the tool.

The Rating Wizard consists of the following components:

- **Criteria:** Sets which asset type or Assets are to be evaluated and rated
- **Rules:** Uses the EOL rules contained in the Asset Type screens to measure against (these can also be adjusted on the fly if desired)
- **Calculated Setup:** an automated process by which assets are selected and updated en masse
- **Manual Setup:** Allows to override asset(s) if there are exceptions

The screenshot shows the 'Rating Wizard' interface. On the left is a blue sidebar with a menu containing: Work Order, Work Request, Asset, Asset Lifecycle (highlighted), Capital Planning, Rating Wizard (highlighted), POF Wizard, COF Setup, Criticality Wizard, Lifecycle, AL Reports, Inventory, and Reporting. The main area has a title 'Rating Wizard' with a refresh icon. Below the title are four tabs: Criteria (selected), Rules, Calculated, and Manual. The 'Criteria' tab contains several input fields: 'Workgroup' (dropdown menu showing 'FACIL'), 'Asset Type*' (dropdown menu with a search icon), 'Asset ID' (dropdown menu with a refresh icon), 'Min Replacement Cost' (text input), 'Age in Years' (text input), and 'Location' (dropdown menu with a search icon). Below these fields are two green buttons: 'Retrieve' and 'Reset All'. Under the buttons, it says 'Press "Retrieve" to see the table'. At the bottom, there is a section titled 'Map Assets' with the text 'No Assets Selected'.

To successfully run the asset wizard, the base EOL and related data must be populated on the assets, as discussed in this document. Furthermore, a default Condition Index MUST be assigned. MaintStar understands it is nearly impossible to have data in every single field for every single asset. As such, it does it's best to take what data is there and come up with the best rating based on the information entered. The following precedence is set within MaintStar when evaluating data:

- Age, then
- Meter (if present), then
- Work Order Cost, then
- Current Rating (if present)

Running the Rating Wizard

1. Navigate to Asset Lifecycle > Asset Rating Wizard
2. Select the Criteria tab
3. Select either a single Asset Type (must be done one at a time) or multiple Asset IDs, click Retrieve

Rating Wizard

Criteria Rules Calculated Manual

Workgroup: FACIL

Asset Type: AM ASSET x

Asset ID:

Min Replacement Cost: Age in Years:

Location:

Retrieve Reset All

Asset ID	Asset Type	Asset Description	Current Rating	Current Age	Remaining Life Years	Useful Life Years	Replacement Cost	Install Date
am0009	AM ASSET		9	0.00	12.00	12.00		09/03/2024
AM001	AM ASSET	Generator	10	5.00	10.00	15.00		07/01/2019
AM002	AM ASSET	Vehicle	10	4.00	11.00	15.00		01/01/2020

4. Assets within the Asset Type will be retrieved and displayed. From here any key data to be used by the wizard will be displayed (Age, rating, etc).
5. Click the Rules tab
6. This displays the default EOL values assigned to the asset type. For purposes of the rating, these values can be adjusted directly from this screen by clicking "Edit"

7. When ready, click “Calculated Setup” (alternatively Manual Setup) can be used and is discussed in the steps below.
8. Click the “Calculate” button. This will model (but not apply) the ratings. The dialog box will display statistics on how many assets will be changed, unchanged, or with insufficient data.

- Click OK
- The preview screen will display a grid of reasons, current rating and recommended rating.

11. Select “All” assets or Select individual assets.
12. **THE FOLLOWING IS IRREVERSIBLE.** When ready, click “Apply” ratings will be applied to asset(s).

Calculate		Apply												
☒ Edit		Asset ID	Asset Type	Asset Description	Current Rating	Proposed Rating	Reason for Change	EOL (years)	Current Age	Remaining Life Years	Current EOL Age %	Useful Life Years	EOL Meter (from Asset Type)	Current Meter
☒ Edit		AM001	AM ASSET	Generator	3	5	EOL Cost	12.00	4.00	8.00	33%	15.00	100,000.00	
☒ Edit		AM002	AM ASSET	Vehicle	3	2	EOL Age	12.00	2.00	10.00	33%	15.00	100,000.00	1.00

13. To use “Manual Setup”, set the Criteria, Rules as defined above.
14. From here a blanket rating can be assigned to assets.

REGARDLESS OF CALCULATED OR MANUAL RATINGS, A CORRESPONDING ASSET RATING WILL BE RECORDED IN THE ASSET ASSESSMENT TAB OF THE ASSETS WHICH HAVE BEEN UPDATED

Once ratings have been recorded, a variety of reports can be executed, for instance End of Life, Condition Rating, etc. The example below shows a Risk Matrix Report showing assets at risk of failing.

AL Reports													
Risk Matrix													
☒ Asset Type ☐ Workgroup Asset Type: AM ASSET x Age in Years: Criteria Print Run													
Asset Management Risk Matrix Report													
Workgroup: FACIL Asset Type: AM ASSET													
?	Probability of Failure	Conseq. of Failure	Criticality	Asset Type	Generic Type	Asset Id	Description	Asset	Age Yrs	Rating	Remain. Service Life Years	Replacement Cost	Useful Life
?	4	3	12	AM ASSET		am0009		☐	0	10.00	12	\$500.00	12
?	3	3	9	AM ASSET		AM001	Generator	☐	5	1.00	7		15
?	4	3	12	AM ASSET		AM002	Vehicle	☐	4	10.00	8		15
?	5	3	15	AM ASSET		AM003	Truck	☐	1	10.00	11		12
?	1	3	3	AM ASSET		AM004		☐		5.00	12		12
?	3	3	9	AM ASSET		AM008	Vehicle	☐	4	4.00	8	\$750.00	15

Rating Wizard with Map

At times, it may be helpful to leverage GIS data to visualize Asset data on a map, specifically when assets are grouped by or impacted by location. As with many other modules within MaintStar, the map can be enabled which uses the GIS data and displays the selected assets on the map.

The screenshot displays the 'Asset Rating Wizard' interface. On the left, there is a form with the following fields: 'Workgroup' (set to SEWER), 'Asset Type' (set to ssMANHOLE), 'Asset ID' (empty), 'Min Replacement Cost' (empty), 'Age in Years' (empty), and 'Location' (empty). Below these fields are 'Retrieve' and 'Reset All' buttons. A table below the form shows a list of assets with columns for Asset ID, Asset Type, Asset Description, Current Rating, and Current Age. The table contains five rows of data. On the right, a map view shows a residential area with numerous brown circular markers representing assets. The map includes street names like 'Dublin Creek', 'Macy's', and 'Murray's'.

Asset ID	Asset Type	Asset Description	Current Rating	Current Age
SA2D3M405	ssMANHOLE	625		
SA2D3M406	ssMANHOLE	1138		
SA2D3M407	ssMANHOLE	1168		
SA2D3M408	ssMANHOLE	1149		
SA2D3M409	ssMANHOLE	1152		

COF Setup, POF Wizard and Criticality Wizard

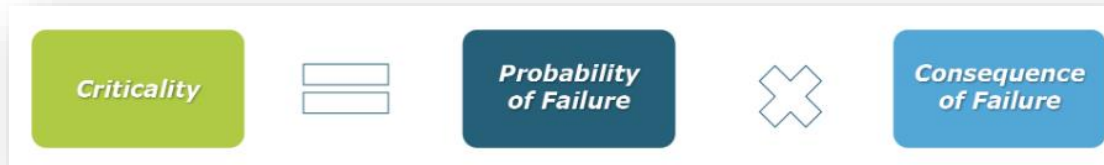
Many organizations wish to apply asset criticality scores to assets to help identify assets of great risk (or criticality). This score provides an additional point of data for lifecycle planning. This score is not to be confused with condition ratings, but rather is a score of risk (or criticality) and possible impact should it fail. As a reminder:

- **Consequence of Failure (COF):** a value judgement of failure using the criteria of financial, environmental, and social impacts which can be valued differently by giving more weight to one type of consequence than another
- **Probability of Failure (POF):** a determination of the likelihood or probability of failure, which is often adjusted due to asset condition, frequency of use, environmental conditions, etc.
- **Criticality:** the measure of risk/importance associated with an asset

As mentioned above, criticality is a risk assessment process. The overall risk is determined by the probability of failure and the consequence of failure. The assets that have the greatest probability of

failure and the greatest consequences due to failure will be the assets that are the highest risk and therefore the most critical. The assets that have low likelihood of failure and low consequences if there is a failure will be the least critical assets.

To calculate the criticality score for a given asset, multiply its probability of failure rating by its consequence of failure rating.



Criticality Scores (without Redundancy)

Most POF and COF values are from 1 to 5, with 1 being a low consequence and probability and 5 being a very high risk consequence and probability of failure. Following is a simple example:

$$POF=2 * COF=4 = \text{Criticality of } 8$$

Consequence (Cost) of Failure	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
	Probability of Failure					
Multiplied		1	2	3	4	5
1 Very Low		2 Low		3 Moderate		4 High
						5 Very High

Criticality Scores (with Redundancy)

In addition to the above, organizations may choose to factor redundant systems into their criticality scores. The idea is that if a backup system exists then the consequence of failure is lowered. The calculation for this would be:

$$\text{Criticality} = POF * COF * (100 - \text{Redundancy}) / 100$$

An example pump with a backup pump would reduce criticality by half, and would look as follows:

POF	2	COF	2
Redundancy	50	Criticality	2

Without redundancy, the criticality would be a 4, but with redundancy, it is now a 2. As a reference, values for backups would use these redundancy factors:

- No backup enter 0 or no value
- Primary plus backup of 1 unit enter 50
- Primary plus backup of 2 units enter 66
- Primary plus backup of 3 units enter 75
- Redundancy cannot be greater than 99

MaintStar offers two methods for assigning criticality values: Manual or via a Wizard. Using the manual option sets the POF and COF to each asset individually. This could be overwhelming with many assets. Both options are discussed below.

Running the Consequence of Failure (COF) Setup

MaintStar provides licensed tools to automate much of the data entry. The idea is to first set the Consequence of Failure (COF) for individual asset types which are then applied directly to the asset type. directly on the asset type screen.

Setting COF Values to Assets using the Setup Tool

1. Navigate to Asset Lifecycle > COF Setup
2. Select the appropriate filters, such as Asset Type(s) – note that multiple types can be selected

COF Wizard

Workgroup Asset Type Parent Asset Parent Asset Id

Consequence of Failure Wizard
Asset Type: AM ASSET

Edit	Asset Type	Asset Count	COF=1	COF=2	COF=3	COF=4	COF=5
☐ Edit	AM ASSET	3	0	0	0	0	3

3. Click Run – this will pull in the selected asset types
4. If the intention is to overwrite ALL COF values for particular Asset Types, click Edit on the Asset Type and update with the desired value.

Apply COF Values to Assets (this is irreversible unless the Wizard is re-run)

1. If the intention is to set or update a new COF value for the chosen Asset Type, Edit the line, enter the “New COF from Asset Type Setup” (1 to 5 value).

COF Wizard

Workgroup: [dropdown] Asset Type: [dropdown] Parent Asset: [dropdown] Parent Asset Id: [dropdown]

AM ASSET x

Run Update Selected Asset Types Print

Save Reset

Consequence of Failure Wizard

Asset Type: AM ASSET 07/23/2024

Cancel	Asset Type	Asset Count	COF=1	COF=2	COF=3	COF=4	COF=5	New COF from Asset Type Setup
☐ Cancel	AM ASSET	3	0	0	0	0	3	2

2. When ready, click Save, then “Update Asset Type”
3. Those values will now be placed on ALL Assets within the type

Running the Probability of Failure (POF) Wizard

MaintStar provides licensed tools to automate much of the data entry. Once the Consequence of Failure (COF) has been identified for an asset type through either manual entry or the COF tool, POFs can be calculated using the POF Wizard.

Setting POF Values to Assets using the Wizard

1. Navigate to Asset Lifecycle > POF Wizard
2. Select an Asset Type, then click Retrieve

POF Wizard

Criteria Rules Calculated Manual

Workgroup: FACIL

Asset Type: [dropdown] AM ASSET x

Asset ID: [dropdown]

Min Replacement Cost: [input] Age in Years: [input]

Location: [dropdown]

Retrieve Reset All

Reset

	Asset ID	Asset Type	Asset Description	Current POF	Current Age	Remaining Life Years	Useful Years
	am0009	AM ASSET		5	0.00	12.00	
	AM001	AM ASSET	Generator	5	5.00	10.00	
	AM002	AM ASSET	Vehicle	5	4.00	11.00	
	AM003	AM ASSET	Truck	5	1.00	11.00	

- Click the Rules tab and validate or update the factors that help to determine POF values. As a reminder, these values are pulled from the Asset Type settings. Changing the values here does NOT reset Asset Type settings. This is helpful for performing adjustments as needed.

POF Wizard

Criteria **Rules** Calculated Manual

Workgroup: FACIL

Asset Type: AM ASSET

Retrieve **Reset All**

Edit	Asset Type	EOL Age Years	EOL Meter	EOL Percent	Condition Index	Default	Description
Edit	AM ASSET	6.00	80,000.00	100.00	1 TO 10	☒	1 is best, 10 is worst.

- Click the Calculated tab.
- Click Calculate to Calculate POF Values based on settings above. It will display current (if they exist) and proposed POF Values.

POF Wizard

Criteria Rules **Calculated** Manual

Workgroup: FACIL

Asset Type: AM ASSET

Calculate **Apply**

New POF Calculation Completed

Total Assets Examined: 14

Assets with POF Unchanged: 12

Assets with POF to be Changed: 2

Assets with No Data to Calculate: 0

Ok

☐	Edit	Asset ID	Asset Type	Asset Description	Current POF	Proposed POF	Reason for Change	EOL (years)	Current Age	Remaining Life Years	Current EOL Age %	Useful Life Years
☒	Edit	am0009	AM ASSET		5	1	EOL Cost	6.00	0.00	12.00		12.00
☐	Edit	AM001	AM ASSET	Generator	5				5.00	10.00		15.00
☐	Edit	AM002	AM ASSET	Vehicle	5				4.00	11.00		15.00
☐	Edit	AM003	AM ASSET	Truck	5				1.00	11.00		12.00
☒	Edit	AM004	AM ASSET		2	1	EOL Cost	6.00	0.00	12.00	113%	12.00

6. Click OK
7. Either Select or Deselect rows for desired changes.
8. Click Apply to commit the POF Values.

Running the Criticality Wizard

MaintStar provides licensed tools to automate much of the data entry. Once the Consequence of Failure (COF) and Probability of Failure (POF) values have been assessed (as identified above), the Criticality Calculation can be ran from the Criticality Wizard.

Calculating Criticality

1. Navigate to Asset Lifecycle > Criticality Wizard
2. Select an Asset Type, then click Retrieve

The screenshot displays the 'Criticality Wizard' interface. On the left is a sidebar with navigation options: Work Order, Work Request, Asset, Asset Lifecycle (selected), Capital Planning, Rating Wizard, POF Wizard, COF Setup, Criticality Wizard (highlighted), Lifecycle, AL Reports, Inventory, Reporting, Work Planning, Purchase Orders, Invoicing, Fleet, Setup, and Security.

The main area is titled 'Criticality Wizard' and has two tabs: 'Criteria' (selected) and 'Preview and Calculate'. Under the 'Criteria' tab, there are several input fields: 'Workgroup' (set to 'FACIL'), 'AssetType' (set to 'AM ASSET'), 'Min Replacement Cost', 'Min Criticality', 'Min Rating', 'Age in years' (with a 'To' field), and 'Header'. Below these fields is the formula:
$$\text{Criticality} = \text{POF} * \text{COF} * (100 - \text{Redundancy}) / 100$$
 and a green 'Retrieve assets' button.

Below the form is a table titled 'Asset Management Criticality Report' with the subtitle 'Model: FACIL Asset Type: AM ASSET'. The table has the following columns: Criticality, POF, COF, Redundancy, Asset ID, Asset Type, Description, Asset, Age Yrs, Rating, and Remain Service Life Yrs. The table contains three rows of data:

Criticality	POF	COF	Redundancy	Asset ID	Asset Type	Description	Asset	Age Yrs	Rating	Remain Service Life Yrs
9.00	1.00	2.00	0.00	am0009	AM ASSET		☐	0.00	9.00	6.00
10.00	5.00	2.00	0.50	AM001	AM ASSET	Generator	☐	5.00	10.00	10.00
9.00	5.00	2.00	0.50	AM002	AM ASSET	Vehicle	☐	4.00	10.00	11.00

3. Click Preview and Calculate
4. This screen will COF and POF values as well as criticality, some or all of which might be incorrectly displayed, which is the purpose of this tool

Criticality Wizard

Criteria **Preview and Calculate**

Calculate Criticality

Asset Management Criticality Report
Model: FACIL Asset Type: AM ASSET

Edit	Criticality	Proposed Criticality	POF	COF	Redundancy	Asset ID	Asset Type	Description	Asset	Age Yrs	Rating	Remain Service Life
Edit	9.00		1.00	2.00	0.00	am0009	AM ASSET		☐	0.00	9.00	
Edit	10.00		5.00	3.00	0.50	AM001	AM ASSET	Generator	☐	5.00	10.00	

5. Click Calculate Criticality, then OK
6. All Criticality scores will now be calculated and committed to Assets.

Manual Assignment of COF, POF and Criticality to Individual Assets

As indicated above, these values can be assigned either through automated tools or via manual data entry. Manual entry is the least desired method as it is time consuming, however, certain situations may necessitate its use.

1. Navigate to Assets > Assets locate the desired asset
2. Locate the field labeled “COF”, “POF” and “Criticality” and set the desired values (from 1 to 5) *
*the “cof”, “pof”, and “Criticality” fields may need to be added to asset screen customization(s)
3. Click Save and confirm the proper calculation is performed.

COF 3

POF 4

Criticality 12

4. Only applies the values to one asset. All assets must be individually updated if this path is used.

Capital Planning

The Capital Planning tool is a separately licensed tool that provides the ability to ‘model’ replacement dates and costs base one, some or all the factors above, such as age, meter, cost, ratings, usage and assessments. Going beyond a fixed or static replacement date, this is particularly helpful for budgeting for current and/or future year replacements based on the factors above.

Running Capital Planning

- 1. Navigate to Asset Lifecycle > Capital Planning
- 2. The Setup tab is where differing values can be used for modeling
- 3. In addition to a “Default” model additional modeling scenarios can be created (discussed below)
- 4. Below is a summary of the Capital Planning Setup

Capital Planning

SetupFiscal CalendarReplacement Schedule by AssetAssessments

Inflation Rate Percent5.00

Asset Threshold10.00

First fiscal Year2024

Number of years12

Meter Lookback Years3

Pct Lookback Years3

Add

Edit	Default	Inflation Rate Percent	Asset Threshold	First Fiscal Year
☐ Edit	☐	5.00	10.00	2024

- **Inflation Rate Percent:** The rate of year over year inflation to provide an adjusted replacement cost
 - **Asset Threshold:** The desired Replacement Cost of an Asset. For instance, if this is set to \$10.00 then it will only return items over that amount.
 - **First Fiscal Year:** Returns the year in which modeling is desired.
 - **Number of Years:** Sets how many years ahead to return replacement costs.
 - **Meter Lookback Years:** Sets number of years into history to project how many miles will be driven each year in the future.
 - **Pct Lookback Years:** Sets how many years into history to project how much the maintenance cost will be each year in the future.
- 5. Once desired values are set for the default profile, click Save
 - 6. Click the Fiscal Calendar tab to set Fiscal Year periods (if necessary)

Capital Planning

Setup **Fiscal Calendar** Replacement Schedule by Asset Assessments

 Add  Add Year

☐ Edit	Year No.	Start Date	End Date	Report Flag	Comments
☐ Edit	2019	07/01/2020	06/30/2021	☐	
☐ Edit	2020	06/30/2019	09/18/2020	☐	Yes
☐ Edit	2021	07/01/2021	06/30/2022	☐	
☐ Edit	2022	07/01/2022	06/30/2023	☐	
☐ Edit	2023	07/01/2023	06/30/2024	☐	
☐ Edit	2024	07/01/2024	06/30/2025	☒	
☐ Edit	2025	07/01/2025	06/30/2026	☒	
☐ Edit	2026	07/02/2026	07/01/2027	☒	
☐ Edit	2027	07/02/2027	07/01/2028	☒	
☐ Edit	2028	07/02/2028	07/01/2029	☒	

- Click Replacement Schedule by Assets
- Select a desired Asset Type
- To ensure all factors are considered, select all check boxes, then click Run

Capital Planning

Setup Fiscal Calendar **Replacement Schedule by Asset** Assessments

Workgroup

 Asset Type

Asset Subtype

 Assets

Purchase Cost From

Thru

☒ Use Assessments

☒ Use Ratings

☒ Use Meter

☒ Use Maint Cost Percent

☒ Use Age

Edit	Asset Type	Asset ID	SubType	Descr	Reason	Repl Date	Repl Cost	2024	2025	2026	2027	2028
Edit	AM ASSET	am0009			Rating: 10 is worst	2024-07-01 14:00:00.000	\$500.00	\$500.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM001		Generator	No Repl Cost	2024-09-23	\$0.00	\$22.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM002		Vehicle	No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM003		Truck	No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM004			No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM008		Vehicle	Rating: 4 Yrs: 7.2	2031-12-07	\$750.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM009			No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM011		Generator	No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM012		New asset	No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM111		AM111 - Sedan	Rating: 10 is worst	2024-07-01 14:00:00.000	\$2500.00	\$2500.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM222			No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AMACQ001			No Repl Cost		\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00

10. This will display projected replacement costs and years based on the common factors contained within this document

Setup of Additional Capital Planning Scenarios/Models

MaintStar allows for additional models to be created whereby replacement schedules based on differing scenarios or models can be created. This is helpful for developing replacement schedules based on budgets, changing replacement requirements, or planning for the future with differing 'start' dates.

1. Navigate to Asset Lifecycle > Capital Planning
2. The Setup tab is where differing values can be used for modeling
3. In the grid, click "Add" and establish new values, for example:

Capital Planning

Setup Fiscal Calendar Replacement Schedule by Asset Assessments

Inflation Rate Percent 5.00
Asset Threshold 10.00
First fiscal Year 2024
Number of years 12
Meter Lookback Years 3
Pct Lookback Years 3

Add Save Reset

	Default	Inflation Rate Percent	Asset Threshold	First Fiscal Year	Number of Years	Meter Lookback Years	Pct Lookback Years
☐ Edit	☐	5.00	10.00	2024	12.00	3.00	3.00
☐ Edit	☐	3.5	8	2027	10	10	7

4. Click Save
5. Click on Replacement Schedule by Asset, select appropriate options (as above) and select desired model to run.

MAINTSTAR
Asset Management

Full menu
Home (admin)
Work Order
Work Request
Asset
Asset Lifecycle
Rating Wizard
COF Setup
POF Wizard
Criticality Wizard
Capital Planning
AL Reports
Inventory

Capital Planning

Setup Fiscal Calendar Replacement Schedule by Asset Assessments

Workgroup FACIL x
Asset Type AM ASSET x
Asset Subtype
Default Model Asset Default
Assets
Purchase Cost From
Thru

☐ Use Assessments
☐ Use Ratings
☐ Use Meter
☒ Use Maint Cost Percent
☐ Use Age

Run

Edit	Asset Type	Asset ID	SubType	Descr	Reason	Repl Date	Repl Cost	2025	2026	2027	2028	2029
Edit	AM ASSET	AM001		Generator	Pct Cost 66/75	2026-03-04	\$200.00	\$200.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM002		Vehicle			\$300.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM003		Truck			\$500.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Edit	AM ASSET	AM004					\$600.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00

Asset Lifecycle (AL) Reports

These can be found in Asset Lifecycle > AL Reports. Below is a summary and function of the reports.

The screenshot shows the 'AL Reports' interface. At the top, there is a title 'AL Reports' followed by a hamburger menu icon. Below the title is a horizontal navigation bar with the following tabs: 'Risk Matrix' (which is underlined), 'Rating', 'Alerts', 'Acquisition', 'Disposal', 'Claims', 'Probability of Failure', 'Consequence Of Failure', and 'End Of Life'. Below the navigation bar, there are two radio buttons: 'Asset Type' (selected) and 'Workgroup'. Below these is a search bar labeled 'Asset Type' with a dropdown arrow and a magnifying glass icon. Below the search bar is a label 'Age in Years' followed by two empty input fields. At the bottom of the form area, there are three green buttons: '- Criteria', 'Print', and 'Run'. Below the buttons, there is a text prompt: 'Press "Run" to see the report'.

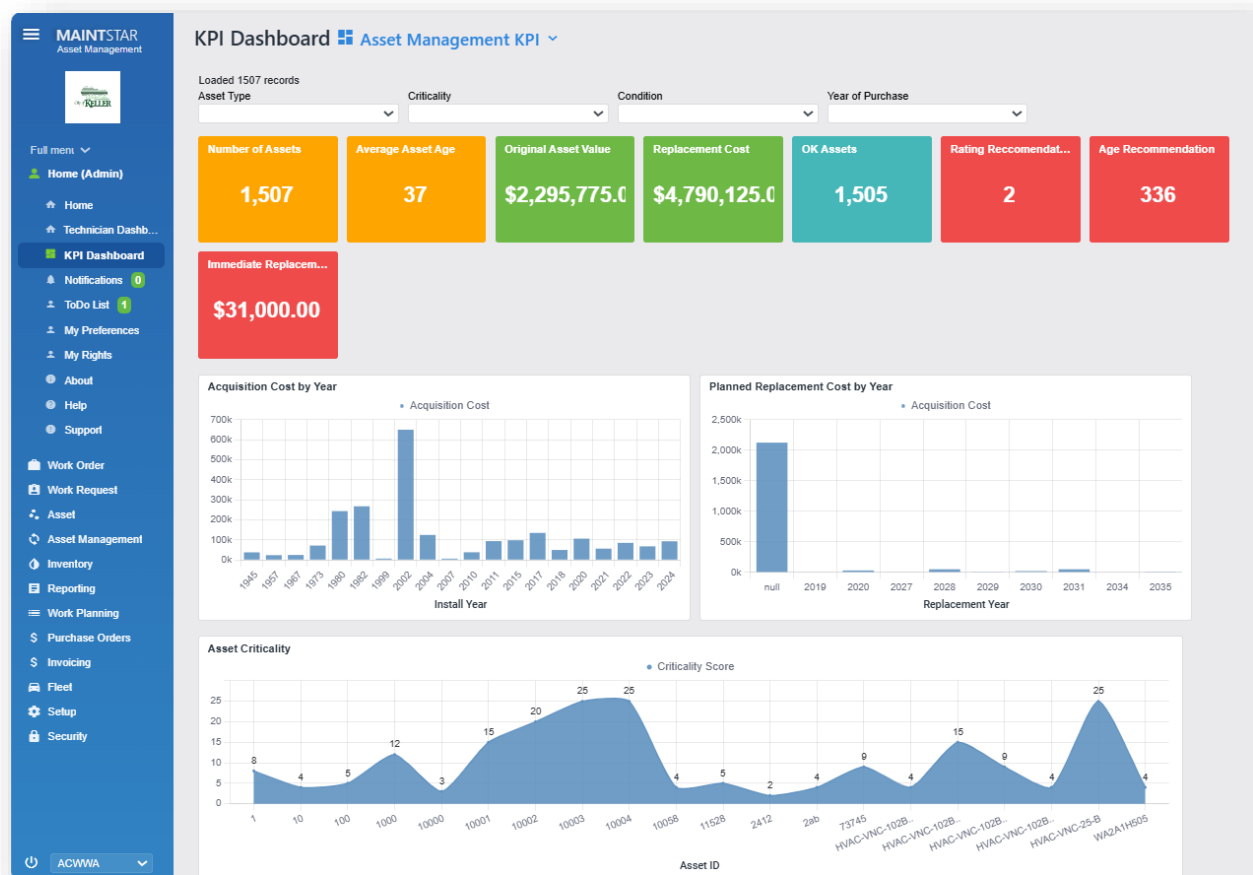
- **Risk Matrix:** This report shows the risk (based on Criticality) of selected assets.
- **Rating:** This report shows a summary of ratings assigned to selected assets.
- **Alerts:** If Alerts are used and entered on Assets, this report shows those Alerts in a single report.
- **Acquisitions:** If the Acquisition tool is used, then this report will show all acquisitions for the desired period.
- **Disposals:** If the Disposal tool is used, then this report will show all disposals for the desired period.
- **Probability of Failure:** This summarizes all POF values for the selected asset type.
- **Consequence of Failure:** This summarizes all COF values for the selected asset type.
- **End of Life:** This report provides a summary of all Assets that are at End of Life.

Additional Reporting Options

As with all modules within MaintStar, custom Queries, Reports and Key Performance Indicators (KPIs) may be extremely helpful in providing results unique to an organization's needs. Below are examples of such reports.

***The KPI module is a separately licensed product**

Example of Asset Report using the KPI Dashboard



Example of Asset Report using the Query Tool

Query 74: PUMP EOL

Details Result Charts

← → 🔄 ⋮

PUMP EOL							
Asset Descr	Asset Id	Asset Type	Install Date	Age Yrs	Remaining Life Yrs ↓	Original Value	Replacement Cost
852	852	wPUMP	05/16/2000	24.00	-14.00	6,000.00	9,000.00
	887766	wPUMP	05/16/2002	22.00	-12.00	500.00	1,300.00
712	712	wPUMP	05/16/2010	14.00	-4.00	5,600.00	9,200.00
5555	5555	wPUMP	05/16/2015	9.00	1.00	9,000.00	12,500.00
							Replacement Cost Total: 32,000.00

PUMP EOL